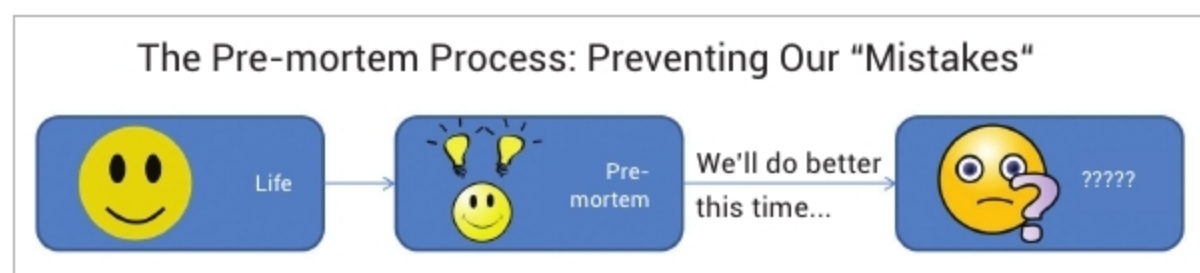


Mitigating AI Risks the Right Way

Managing them systematically and pragmatically is the key to handling artificial intelligence (AI) risks. The problem with AI risks is that these are highly scalable and can quickly grow out of control due to the power of automation and the sheer capacity of AI to execute tasks.



(Image Source: <http://thekaizone.com/wp-content>)

Risk mitigation can become a daunting task if there is no quantifiable measure attached to it. Following the maxim—if you can measure it, you can improve it—a systematic and measurable approach can come in handy.

The method explained in this article can help in establishing quantitative measures. Once you establish those measures, it can become easier to assess the risk and progress of mitigation activities.

The core principles

In the case of artificial intelligence (AI) risks, three core principles serve as a foundation. Generally speaking, these principles are essential for the effective management of risks of all kinds.

Identify risks: While using a systematic approach, one must determine the most severe risks. Without identification there cannot be any control or management. A good cross-section of the team is necessary to work on the risk identification process.

Be comprehensive: AI systems are not one-point solutions. These often involve several other systems, and the final system is pretty much a system of systems. This means that, to be effective, merely covering AI solution is not going to be helpful. You will have to be comprehensive in the risk identification process. Without being comprehensive, risks from other parts of the system can fall through the cracks and wipe out all the efforts.

Be specific: It is not difficult to design and implement AI solutions in a responsible manner. However, to do that, it is a prerequisite to understand risks better and specifically to your application. A broad-based risk identification and management approach is not helpful. Moreover, establishing a control system across the value chain is also quite important.

Pre-mortem analysis

The term project pre-mortem first appeared in the HBR article written by Gary Klein in September 2007. As Gary writes, “A pre-mortem is the hypothetical opposite of a post-mortem.”

It is a project management strategy in which a project team imagines a project failure and works backward to determine what potentially could lead to that failure. This working is then used to handle risks upfront.

However, in the risk management context, we are going to use this (pre-mortem) term interchangeably to represent a more sophisticated and engineering-oriented methodology known as Failure Mode and Effects Analysis (FMEA).

Key elements of the pre-mortem analysis

Pre-mortem analysis or FMEA is typically done by a cross-functional team of subject matter experts (SMEs). A better format to conduct this exercise is in the form of a workshop.

During the workshop, the team thoroughly analyses the design and the processes are implemented or changed. The primary objective is to find weaknesses and risks associated with every aspect of the product, process, or both. Once you identify these risks, take actions to control and mitigate them, and verify that everything is in control.

Pre-mortem analysis record has sixteen columns, as explained below (see figure on next page).

Process step or system function:

This column briefly outlines the function, step, or an item you are analysing. In a multi-step process or multi-function system, there would be several rows, each outlining those steps.

Failure mode: For each step listed in column #1, you can identify one or more failure modes. It is an answer to the question: In what ways can the process or solution fail? Or, what can go wrong?

Failure effects: In case of a failure, as identified in column #2, what are its effects? How can the failure affect key process measures, product specifications, customer requirements, or customer experiences?

